

Lesson 6: Exercises and complete solutions

Eigenvalues, Eigenvectors, and Spectral Intuition

Mbaye Kebe | SRAI - Statistics, Reasoning and Artificial Intelligence

Book 1: Mathematical Foundations | PU-B01-C06 | v0.1.0 | Review candidate

How to use this companion

Attempt the ten exercises before reading the solutions. Show the algebra, state the assumptions, and interpret the result in words. Use the tested Lesson 6 notebook to check calculations, not as a substitute for explaining them. This standalone companion expands the solutions in the approved chapter without changing its examples.

Prerequisites: matrix multiplication, determinants, bases, orthogonality and norms. All data and sector coefficients are synthetic teaching examples. These exercises do not establish empirical policy recommendations.

Notation: a subscript is not a power

For a real vector v , the Euclidean norm is $\|v\|_2 = \sqrt{\sum_i v_i^2}$. Its square is $\|v\|_2^2 = \sum_i v_i^2$. Thus for $v = (3, 4)^T$, the norm is 5 and its square is 25. The subscript identifies the norm; the superscript indicates squaring. For a matrix, $\|A\|_2$ denotes its largest singular value, not generally its largest eigenvalue magnitude.

Exercise 1 - Eigenpairs and direction

For $A = \text{diag}(-2, 0)$, identify two eigenpairs, verify each by multiplication and interpret the transformation. Explain why the zero vector is excluded from the definition of an eigenvector.

Exercise 2 - An orthonormal eigenbasis

For $S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, derive the characteristic polynomial, calculate the eigenvalues and find orthonormal eigenvectors. Verify $Q^T Q = I$ and $S = Q D Q^T$.

Exercise 3 - Powers through an eigenbasis

Using the same S , derive a formula for S^k for integers $k \geq 0$. Calculate S^{10} and verify what the formula gives when $k = 0$.

Exercise 4 - A defective matrix

For $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, prove $J^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ for integers $k \geq 0$. Explain why the matrix cannot be diagonalized.

Exercises 5-10

Exercise 5 - A misleading convergence signal

Let $A = \text{diag}(1, -1)$ and start normalized power iteration at $x_0 = (1, 1)^T / \sqrt{2}$. Calculate the iterates and their Rayleigh quotients. Explain why an unchanged quotient does not establish an eigenpair.

Exercise 6 - A hidden unstable mode

For $U = \text{diag}(1.1, 0.9)$, give a nonzero starting vector whose trajectory tends to zero. Contrast this with a starting vector that grows. What can one decaying trajectory establish about stability for all initial states?

Exercise 7 - Transient amplification

For $T = \begin{pmatrix} 0.8 & 4 \\ 0 & 0.8 \end{pmatrix}$, calculate the spectral radius and derive T^k . With $x_0 = (0, 1)^T$, show that the state can initially grow even though it eventually tends to zero.

Exercise 8 - Covariance and PCA

The four observations, in rows, are

$$X = \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 2.5 \\ 5 & 4 \end{pmatrix}.$$

Center both columns, form the sample covariance matrix, compute its eigenvalues and explained-variance ratios, and explain the denominator $n - 1$. Does a high first-component ratio prove predictive accuracy?

Exercise 9 - What the sector weights mean

Use $x_{t+1} = M x_t$, with rows and columns ordered Agriculture, Health, Energy, Transport:

$$M = \begin{pmatrix} 0.70 & 0.10 & 0.20 & 0.15 \\ 0.10 & 0.80 & 0.25 & 0.10 \\ 0.25 & 0.10 & 0.75 & 0.30 \\ 0.20 & 0.15 & 0.35 & 0.70 \end{pmatrix}.$$

Interpret its dominant eigenvalue, approximately 1.31148205, and its positive eigenvector normalized to sum to one: $(0.20024812, 0.23284988, 0.28255681, 0.28434520)^T$. Check whether M is stochastic. Give three reasons these components cannot prescribe expenditure shares.

Exercise 10 - Accuracy versus dominance

For unit u , consider $r = \|Au - \lambda u\|_2 / (\|A\|_2 + |\lambda|)$, using the numerator alone if the denominator is zero. Explain what a small residual establishes and what it does not. Give an exact nondominant eigenpair with zero residual.

Worked solutions 1-2

Solution 1 - Verify, then interpret

For $e_1 = (1, 0)^T$, direct multiplication gives $Ae_1 = (-2, 0)^T = -2e_1$. Therefore $(-2, e_1)$ is an eigenpair. The direction is reversed and the length is multiplied by two.

For $e_2 = (0, 1)^T$, $Ae_2 = (0, 0)^T = 0e_2$. Thus $(0, e_2)$ is an eigenpair. A zero eigenvalue is permitted: the nonzero eigenvector is mapped to zero. It is the eigenvector, not the eigenvalue, that must be nonzero.

If $v = 0$ were allowed, $Av = \lambda v$ would hold for every scalar λ . It would identify no direction and make the eigenvalue condition uninformative. Any nonzero multiple of an eigenvector remains an eigenvector for the same eigenvalue.

Check: the two coordinate vectors are independent and span the plane. Do not mistake the zero output of Ae_2 for an invalid eigenvector input.

Solution 2 - Derive the decomposition

First expand the determinant:

$$\det(S - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1).$$

The roots are 3 and 1. At $\lambda = 3$, $(S - 3I)v = 0$ gives $v_2 = v_1$, so choose $(1, 1)^T$. At $\lambda = 1$, $(S - I)v = 0$ gives $v_2 = -v_1$, so choose $(1, -1)^T$.

Each has length $\sqrt{2}$. Dividing by that length gives

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, Q = (q_1 \ q_2), D = \text{diag}(3, 1).$$

The dot product $q_1^T q_2 = (1 - 1)/2 = 0$, and each squared norm is one. Hence $Q^T Q = I$ and $Q^{-1} = Q^T$. Since $S q_1 = 3 q_1$ and $S q_2 = q_2$, $S Q = Q D$; multiplying on the right by Q^T proves $S = Q D Q^T$.

Interpretation: coordinates along q_1 are multiplied by three; coordinates along q_2 are unchanged. Symmetry permits an orthonormal eigenbasis. This conclusion does not extend to every real matrix.

Common pitfall: numerical software may return either eigenvector with its sign reversed. This is not an error; verify the eigenpair and match each column to its eigenvalue.

Worked solutions 3-4

Solution 3 - Compute the power without repeated multiplication

Because $Q^T Q = I$, the intermediate factors cancel in repeated products, giving $S^k = Q D^k Q^T$. The eigenvalues are raised to the power, while the eigenvectors stay fixed:

$$S^k = \frac{1}{2} \begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

With $3^{10} = 59049$, the diagonal entries are $(59049 + 1)/2 = 29525$ and the off-diagonal entries are $(59049 - 1)/2 = 29524$. Therefore

$$S^{10} = \begin{pmatrix} 29525 & 29524 \\ 29524 & 29525 \end{pmatrix}.$$

At $k = 0$, $3^0 = 1$, so the formula gives I , as required. At $k = 1$, it gives the original S . These simple checks catch missing factors of one-half and swapped signs.

Interpretation: a component in the first eigendirection grows like 3^k ; the other stays constant. An initial vector entirely in the second eigendirection does not acquire the growing component under exact arithmetic.

Solution 4 - Growth without a complete eigenbasis

Write $J = I + N$, where $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Multiplication shows $N^2 = 0$. Since I commutes with N , the binomial expansion applies and all terms containing N^2 or higher powers vanish:

$$J^k = (I + N)^k = I + k N = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

This includes $k=0$. The characteristic polynomial is $(1-\lambda)^2$, so the only eigenvalue is one, with algebraic multiplicity two. But $(J-I)v=0$ requires $v_2=0$. Its eigenvectors span only the one-dimensional space generated by $(1,0)^T$.

A two-dimensional diagonalization needs two independent eigenvectors. There is only one independent eigendirection here, so no invertible eigenvector matrix exists.

Interpretation: $J^k(0,1)^T = (k,1)^T$ grows even though both eigenvalues equal one. At spectral radius one, eigenvalues alone do not settle boundedness: Jordan structure matters. This does not contradict the strict condition $\rho(A) < 1$ for decay of every state in a finite-dimensional autonomous discrete-time linear system.

Worked solutions 5-7

Solution 5 - The quotient can stop changing too early

Multiplication flips the second coordinate without changing the length. Thus $x_k = (1, (-1)^k)^T / \sqrt{2}$. The sequence alternates between two distinct directions. For every iterate,

$$x_k^T A x_k = 1/2(1-1) = 0.$$

Yet zero is not an eigenvalue of A . Using the proposed $\lambda=0$, $\|Ax_k - 0x_k\|_2 = 1$. Since $\|A\|_2 = 1$, the scaled residual is also 1, not small. Equal successive Rayleigh quotients therefore do not certify convergence. Use a residual-based check, an iteration limit and justified spectral assumptions.

Solution 6 - One trajectory can conceal instability

Choose $x_0 = e_2$. Then $U^k x_0 = (0, 0.9^k)^T \rightarrow 0$. In contrast, starting at e_1 gives $(1.1^k, 0)^T$, which grows. The spectral radius is 1.1, so the system does not decay for every initial state.

Even $x_0 = (\varepsilon, 1)^T$ with very small nonzero ε ultimately develops a growing first component: $(\varepsilon 1.1^k, 0.9^k)^T$. A short simulation can miss that behavior. One successful trajectory demonstrates only what happened for that start and time horizon.

Solution 7 - Eventual decay need not be monotone

The triangular matrix has eigenvalues 0.8 and 0.8. Thus $\rho(T) = 0.8 < 1$. Write $T = 0.8I + 4N$ using the same nilpotent N as in Solution 4. For $k \geq 1$,

$$T^k = \begin{pmatrix} 0.8^k & 4k(0.8)^{k-1} \\ 0 & 0.8^k \end{pmatrix}.$$

With $x_0 = e_2$, the initial norm is 1. After one step $x_1 = (4, 0.8)^T$, whose norm is $\sqrt{16.64} \approx 4.0792$. At step two $x_2 = (6.4, 0.64)^T$. The coupling creates substantial transient amplification.

Nevertheless, both 0.8^k and $k(0.8)^{k-1}$ tend to zero, so every state eventually decays. **Practical lesson:** a long-run stability claim does not guarantee acceptable short-run losses or loads. Examine trajectories, time horizons and operating limits as well as spectral radius.

Worked solution 8 - Covariance and PCA

Center the observations

The column means are $\bar{x} = (3.5, 2.375)$. Subtract them from each row:

$$X_c = \begin{pmatrix} -1.5 & -1.375 \\ -0.5 & -0.375 \\ 0.5 & 0.125 \\ 1.5 & 1.625 \end{pmatrix}.$$

The first centered sum of squares is 5; the second is 4.6875; the cross-product sum is 4.75. Since $n = 4$, divide by 3:

$$C = \frac{X_c^T X_c}{3} = \begin{pmatrix} 5/3 & 19/12 \\ 19/12 & 25/16 \end{pmatrix}.$$

Calculate the eigenvalues

The trace is $155/48$ and the determinant is $7/72$. For a two-by-two matrix, the roots satisfy $\lambda^2 - \text{tr}(C)\lambda + \det(C) = 0$. Substitution yields

$$\lambda_{1,2} = \frac{155 \pm \sqrt{23129}}{96},$$

or approximately 3.19877307 and 0.03039360. Their sum equals the trace and their product equals the determinant; these are independent arithmetic checks. Both are positive, as expected for this full-rank sample covariance matrix.

Interpret the proportions carefully

Divide each eigenvalue by the total variance $155/48$. The proportions are approximately 0.99058779 and 0.00941221, or 99.0588% and 0.9412%. The first principal component retains almost all of the variance in these four centered observations, using the supplied units.

For a unit covariance eigenvector q , the score column is $X_c q$, and its sample variance is $q^T C q = \lambda$. This connects the algebra to the statistical interpretation.

Why divide by n minus one?

Estimating the mean from the sample imposes a centered-sum constraint, leaving $n - 1$ degrees of freedom for each variable. Under independent, identically distributed sampling with finite second moments, this correction makes the sample covariance unbiased for population covariance. It does not cure selection bias, dependence or measurement error.

Limits: four synthetic observations do not establish reliable population structure or predictive performance. PCA maximizes retained variance, not causal relevance. Rescaling one variable changes covariance PCA; standardizing is a separate modeling choice that must be disclosed. A high variance ratio is not an out-of-sample accuracy score.

Worked solutions 9-10

Solution 9 - A model direction, not a spending formula

Under the stated convention, M_{ij} maps component j into next-period component i . Multiplying M by the reported positive vector gives approximately 1.31148205 times that vector. Along this direction the model scales by about 31.15% per step. The time step and units are hypothetical; this is not a forecast of annual economic growth.

Row sums are $(1.15, 1.25, 1.40, 1.40)$ and column sums are $(1.25, 1.15, 1.55, 1.25)$. Neither set is all ones, so the matrix is neither row-stochastic nor column-stochastic. Normalizing an eigenvector does not turn the matrix into a transition-probability model.

The displayed components sum to approximately one (rounding causes a tiny discrepancy). Energy and Transport have the larger components of this particular mode. Their values are not optimal budget shares, probabilities of success or estimated causal effects.

Three separate missing ingredients prevent an allocation recommendation: the coefficients are invented rather than validated against observations; interventions are not causally identified; and the model contains no explicit objective, costs, budget constraints or distributional requirements. A fourth concern is sensitivity to uncertain coefficients and an uncertain time horizon.

Defensible conclusion: this specified synthetic linear system has an amplifying dominant mode. Further empirical and decision analysis would be needed before operational use.

Solution 10 - A correct eigenpair may not be dominant

The numerator measures the failure of the eigenpair equation for a unit vector. Scaling by matrix and eigenvalue magnitudes makes the residual easier to compare across examples. A small value supports numerical consistency with an approximate eigenpair; it does not, alone, prove dominance or small eigenvector error for an ill-conditioned problem.

Take $A = \text{diag}(3, 1)$, $u = e_2$ and $\lambda = 1$. Then $Au - \lambda u = 0$, so $r = 0$. Nevertheless, the spectral radius is 3 and this eigenpair is nondominant. Power iteration started exactly at e_2 can remain there.

For these small examples, compute all eigenvalues with a dense eigensolver and compare magnitudes, then check the residual separately. For larger problems, use justified convergence assumptions or an appropriate spectral method. A unique largest modulus and a suitable initial component are important assumptions, not automatic facts.

If the denominator is zero, $A = 0$ and $\lambda = 0$; using the numerator avoids division by zero. Always reject a zero candidate vector before normalization. A successful arithmetic check is not evidence that the underlying model represents reality.

Laboratory verification and review

Work from the tested notebook

Use **M1_N06_eigenvalues_eigenvectors_spectral_intuition_v0.1.0.ipynb** and its matching dependency package. Start a fresh kernel or Colab runtime and run all cells in order, including setup and the final checks. Do not replace the bundled `srai_math` package, change its hashes, skip assertions or disable checks merely to obtain a pass.

The notebook's explicit reference iteration and residual safeguards address limitations in the bundled prerelease runtime. A generic call to another power-iteration routine is not an equivalent acceptance test. Preserve the versioned dependency instructions.

Independent arithmetic cross-check

The following optional cell uses NumPy already available in the notebook environment. It does not replace the notebook's `srai_math` checks.

```
import numpy as np
S = np.array([[2., 1.], [1., 2.]])
assert np.allclose(np.linalg.matrix_power(S, 10),
                   [[29525, 29524], [29524, 29525]])
X = np.array([[2., 1.], [3., 2.], [4., 2.5], [5., 4.]])
C = np.cov(X, rowvar=False, ddof=1)
assert np.allclose(C, [[5/3, 19/12], [19/12, 25/16]])
assert np.allclose(np.linalg.eigvalsh(C),
                   [0.03039360, 3.19877307])
print("Independent arithmetic checks: PASS")
```

Expected output: **Independent arithmetic checks: PASS**. An assertion failure means stop and compare the inputs, centering, denominator and operation with the stated problem. For a missing NumPy import, select the intended kernel and rerun the notebook's approved setup. Do not install an unrelated package named `srai_math`.

What to retain

Keep your written derivations, the executed notebook, its filename/version, Python and package versions, and the complete final test output. Check that formulas and figures render correctly, not only that code executes. Never include passwords, environment-file contents or database backups in the submission.

Completion standard

A complete response gives the ten answers with reasoning, distinguishes norm index from squaring, explains the two convergence counterexamples, and states why the sector weights are not allocations. Report failed checks honestly. The previously reported VS Code and Colab passes concern the notebook; they do not constitute independent review of this new companion.

Source: approved Lesson 6 chapter v0.1.0, sections 6.3-6.9, and its matching notebook, sections 2-9. This companion is a new review candidate. The approved chapter and tested notebook have not been modified; companion approval and publication authorization remain pending.